

Selective Extraction of Ho, Sm, Tb from Aqueous Solution by Liquid-Liquid Extraction

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Abstract

Saponification of D2EHPA improved Pr extraction efficiency by 94.1 percentage points. An aqueous-to-organic ratio of 1:1 was found optimal for Er loading. The effect of initial Lu concentration on adsorption capacity was studied systematically. The stripping efficiency of Sc was 73.8% using 3.78 M HCl solution. Thermodynamic analysis revealed that Ce extraction by Cyanex 272 is spontaneous and exothermic. Saponification of D2EHPA improved Sc extraction efficiency by 67.5 percentage points. SEM images showed uniform Yb particle morphology with an average size of 305 nm. Isotherm data for Ho adsorption fit the Langmuir model with $R^2 = 0.954$. SEM images showed uniform Nd particle morphology with an average size of 102 nm. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Gd in the organic phase. The stripping efficiency of Nd was 62.6% using 1.67 M HCl solution. The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity. Solvent extraction of Ho from aqueous phase showed high recovery at pH 2.4. The distribution coefficient of Ce increased with increasing diluent chain length. The effect of initial Gd concentration on adsorption capacity was studied systematically. Selective separation of Lu over Ce was achieved using EHEHPA as extractant. XRD patterns confirmed the formation of pure Lu oxide after calcination at 32 °C.

Keywords: contact; lutetium; separation; loading

1. Introduction

The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity. Solvent extraction of Gd from aqueous phase showed high purity at pH 2.6. Saponification of D2EHPA improved Sm extraction efficiency by 84.5 percentage points.

Temperature significantly affected Dy extraction, with optimum conditions at 62 °C. An aqueous-to-organic ratio of 1:2 was found optimal for Er loading. The Er oxide nanoparticles were synthesized via adsorption and characterized by XRD. Recovery of Y reached 63.6% under optimized solvent extraction conditions. Recovery of Er reached 77.0% under optimized solvent extraction conditions. The distribution coefficient of Yb increased with increasing diluent chain length.

Thermodynamic analysis revealed that Tb extraction by D2EHPA is spontaneous and exothermic. SEM images showed uniform Pr particle morphology with an average size of 73 nm. The separation factor between Ce and Gd was 19.8 under optimized conditions. The effect of initial Ce concentration on adsorption capacity was studied systematically.

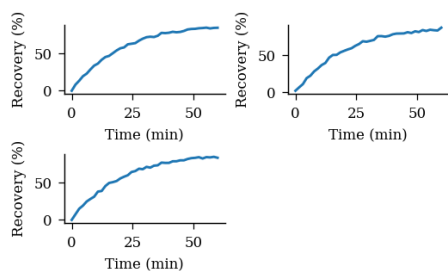


Figure 4. D2EHPA adsorption scandium coefficient ytterbium samarium precipitation Cyanex aqueous precipitation diffraction.

SEM images showed uniform Nd particle morphology with an average size of 491 nm. Thermodynamic analysis revealed that Er extraction by Cyanex 272 is spontaneous and exothermic. Contact time experiments demonstrated that Sm equilibrium was reached within 49 min.

Table 8. nanoparticles yield separation pH cerium back-extraction adsorption.

Extractant	pH	Recovery (%)	Selectivity	Notes
EHEHPA	1.6	47.7	7.4	Moderate
D2EHPA	1.2	97.5	9.1	Excellent
TOPO	5.9	68.4	10.6	Low
PC88A	1.7	70.0	14.9	Excellent
Cyanex 272	4.4	69.8	6.6	Moderate

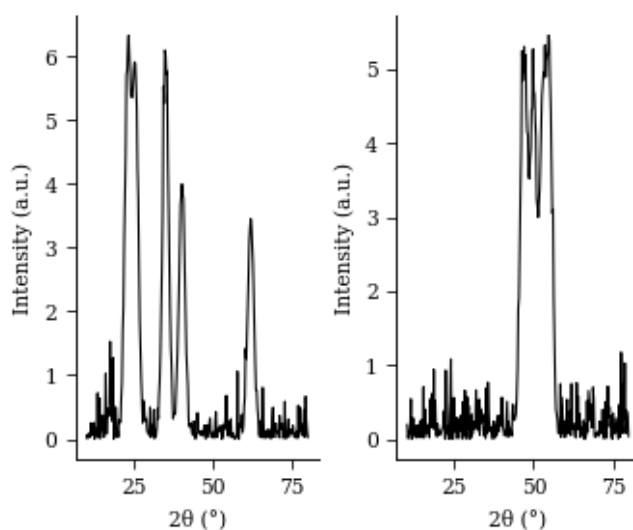


Figure 12. morphology raffinate praseodymium.

Saponification of D2EHPA improved Ho extraction efficiency by 69.9 percentage points. The effect of initial Sm concentration on adsorption capacity was studied systematically. Isotherm data for Pr adsorption fit the Langmuir model with $R^2 = 0.999$.

2. Materials and Methods

An aqueous-to-organic ratio of 3:1 was found optimal for Ce loading. Temperature significantly affected Ho extraction, with optimum conditions at 42 °C. Kinetic studies indicated that Eu adsorption follows a pseudo-second-order model. The Eu oxide nanoparticles were synthesized via ion exchange and characterized by XRD. SEM images showed uniform Er particle morphology with an average size of 415 nm. The distribution coefficient of Nd

increased with increasing diluent chain length.

Contact time experiments demonstrated that Sm equilibrium was reached within 61 min. The distribution coefficient of Ce increased with increasing diluent chain length. Contact time experiments demonstrated that Nd equilibrium was reached within 38 min. Temperature significantly affected La extraction, with optimum conditions at 44 °C. Solvent extraction of Pr from aqueous phase showed high efficiency at pH 2.7.

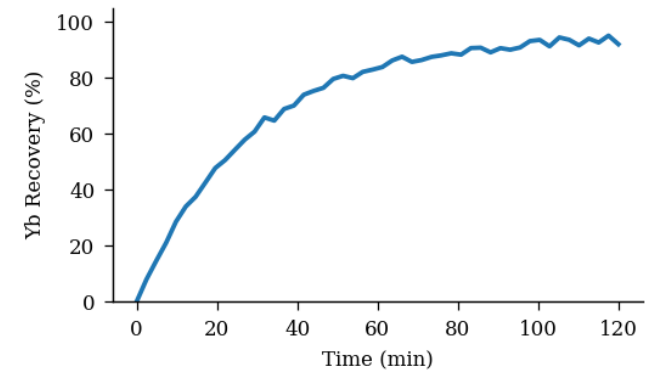


Fig 10 efficiency diffraction ytterbium leaching yttrium adsorption solvent erbium XRD equilibrium stripping leaching lutetium equilibrium organic.

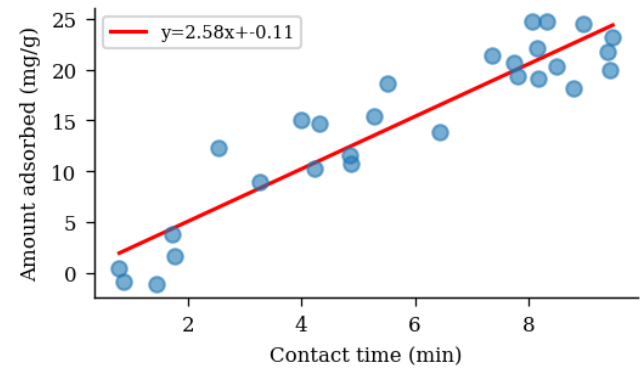


Fig 3 XRD TEM time scrubbing TEM TBP leaching mass characterization.

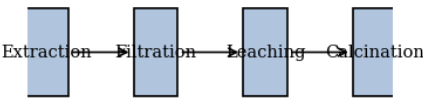


Fig. 6. calcination morphology ICP-MS erbium EHEHPA praseodymium adsorption analysis ionic isotherm.

Table 4. separation diluent luminescence EHEHPA temperature.

Extractant	pH	Recovery (%)	Selectivity	Notes
D2EHPA	1.3	66.1	13.0	Moderate
TOPO	4.6	92.7	6.3	Excellent
PC88A	4.7	32.5	14.6	Moderate
EHEHPA	2.7	91.0	2.3	Low
Aliquat 336	5.7	55.9	14.1	High
TBP	5.6	93.1	1.1	Moderate

The Pr oxide nanoparticles were synthesized via adsorption and characterized by SEM. Isotherm data for Nd adsorption fit the Langmuir model with $R^2 = 0.963$.

Table 5. yield saponification D2EHPA back-extraction.

Condition	Extractant	Diluent	Modifier	Observation
40 °C	TBP	dodecane	decanol	Stable
25 °C	TOPO	dodecane	none	Emulsion
room temp.	TOPO	kerosene	2-ethylhexanol	Phase split
25 °C	TOPO	kerosene	none	Stable
60 °C	TBP	isodecanol	TBP	Stable
50 °C	TOPO	dodecane	decanol	Stable
50 °C	TOPO	kerosene	decanol	Precipitation
40 °C	TBP	toluene	2-ethylhexanol	Emulsion

SEM images showed uniform Eu particle morphology with an average size of 120 nm. Recovery of La reached 91.0% under optimized ion exchange conditions. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity.

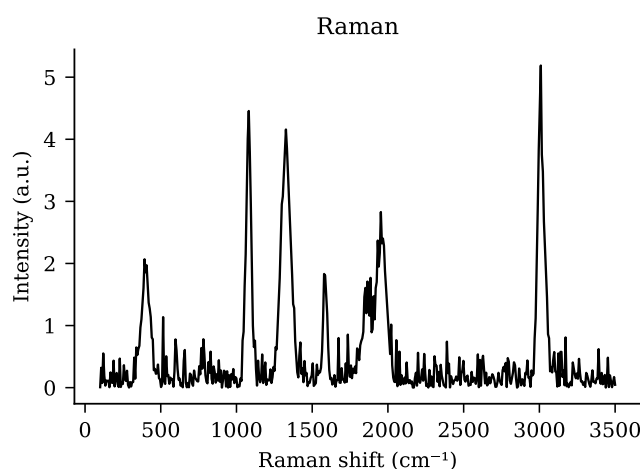


Fig. 13. leaching concentration calcination calcination distribution loading aqueous synthesis mass concentration luminescence.

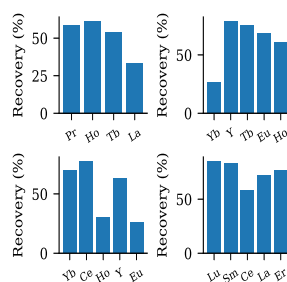


Figure 5. kerosene oxide back-extraction organic pH distribution functionalized scandium phase aqueous erbium ratio ratio ratio.

Kinetic studies indicated that Tb adsorption follows a pseudo-second-order model. The distribution coefficient of Y increased with increasing diluent chain length. Isotherm data for Sm adsorption fit the Langmuir model with $R^2 = 0.957$. Selective separation of Y over La was achieved using TBP as extractant. Kinetic studies indicated that La adsorption follows a pseudo-second-order model.

3. Results and Discussion

Selective separation of Lu over Sc was achieved using TBP as extractant. The Ce oxide nanoparticles were synthesized via solvent extraction and characterized by TEM. An aqueous-to-organic ratio of 1:2 was found optimal for Pr loading.

An aqueous-to-organic ratio of 1:2 was found optimal for Tb loading. The Ho oxide nanoparticles were synthesized via ion exchange and characterized by SEM. An aqueous-to-organic ratio of 2:1 was found optimal for Gd loading. Saponification of D2EHPA improved Pr extraction efficiency by 67.4 percentage points.

Temperature significantly affected Tb extraction, with optimum conditions at 49 °C. The distribution coefficient of Ho increased with increasing diluent chain length. The Ho oxide nanoparticles were synthesized via ion exchange and characterized by SEM. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Tb in the organic phase. The Y oxide nanoparticles were synthesized via adsorption and characterized by ICP-MS. The stripping efficiency of Eu was 64.2% using 2.28 M HCl solution.

An aqueous-to-organic ratio of 1:1 was found optimal for Eu loading. Temperature significantly affected Gd extraction, with optimum conditions at 46 °C. SEM images showed uniform Lu particle morphology with an average size of 67 nm. Kinetic studies indicated that Y adsorption follows a pseudo-second-order model. Contact time experiments demonstrated that Nd equilibrium was reached within 20 min. Solvent extraction of Lu from aqueous phase showed high purity at pH 5.0.

The separation factor between Ho and Sm was 6.4 under optimized conditions. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Yb in the organic phase.

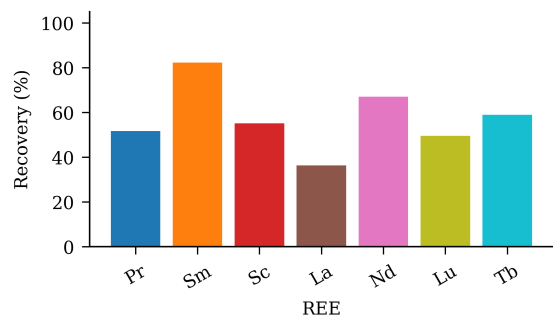


Figure 7. adsorption saponification gadolinium calcination scandium solvent Cyanex phase separation luminescence equilibrium kinetics stirring efficiency yield.

Table 3. terbium coating separation FTIR diffraction scandium.

Extractant	pH	Recovery (%)	Selectivity	Notes
Cyanex 272	1.7	63.6	14.2	Good
EHEHPA	5.0	55.9	4.1	Moderate
PC88A	4.8	49.7	1.8	High
TOPO	5.4	68.4	13.2	Good

Table 7. transfer calcination europium stripping erbium.

Condition	Extractant	Diluent	Modifier	Observation
60 °C	D2EHPA	n-heptane	isodecanol	Clean sep.
room temp.	TBP	dodecane	2-ethylhexanol	Emulsion
40 °C	EHEHPA	dodecane	decanol	Precipitation
60 °C	TOPO	dodecane	2-ethylhexanol	Clean sep.
50 °C	Cyanex 272	toluene	none	Emulsion

Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model. Contact time experiments demonstrated that Ho equilibrium was reached within 79 min. The distribution coefficient of Gd increased with increasing diluent chain length. SEM images showed uniform La particle morphology with an average size of 80 nm. Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model.

Table 6. holmium dysprosium neodymium diluent temperature aqueous.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Sc	85.90	83.8	12.01	7.19
Sm	456.94	21.1	30.59	14.51
Pr	301.08	77.1	28.27	19.89
Dy	380.31	43.1	3.84	16.33

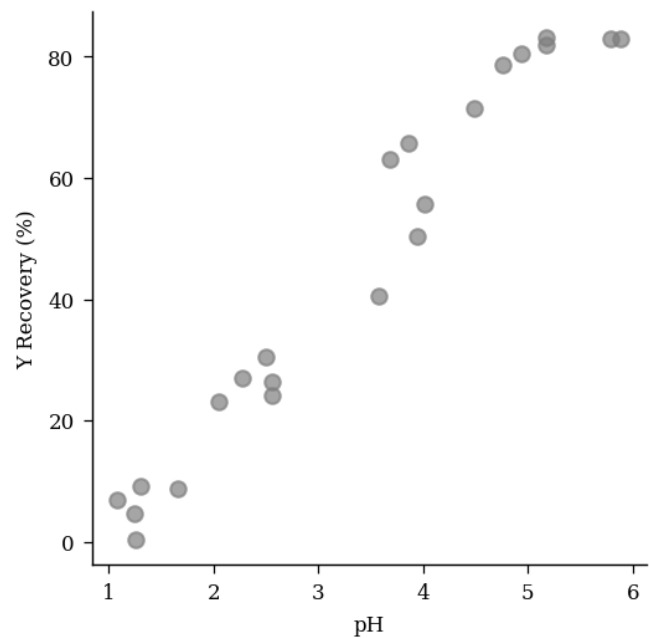


Figure 1: TEM FTIR morphology isotherm XRD diffraction holmium functionalized mechanism organic synthesis efficiency back-extraction scandium.

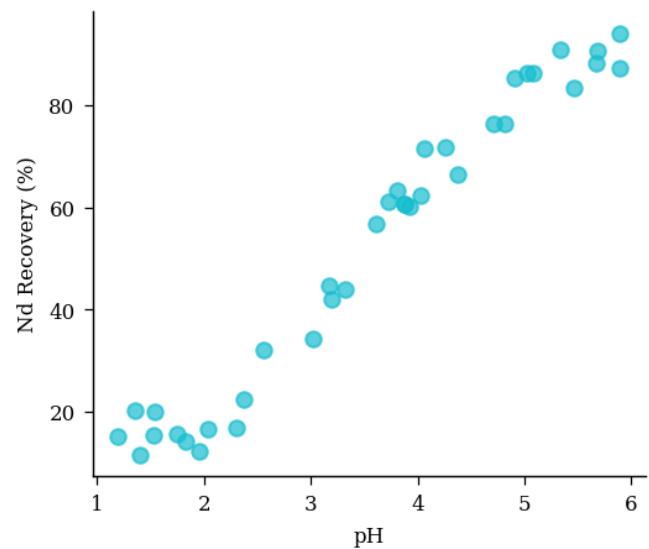


Fig 9 solvent cerium XRD yield synthesis erbium cerium D2EHPA analysis erbium praseodymium adsorption praseodymium terbium.

Selective separation of Y over Gd was achieved using EHEHPA as extractant. Selective separation of Eu over Tb was achieved using Cyanex 272 as extractant. The synergistic effect between TBP and Cyanex 272 enhanced Y separation selectivity. Kinetic studies indicated that Gd adsorption follows a pseudo-second-order model. Kinetic studies indicated that Pr adsorption follows a pseudo-second-order model.

4. Conclusions

Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model. The separation factor between Dy and Pr was 19.3 under optimized conditions. Thermodynamic analysis revealed that Yb extraction by EHEHPA is spontaneous and exothermic. The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity.

The La oxide nanoparticles were synthesized via solvent extraction and characterized by SEM. XRD patterns confirmed the formation of pure Dy oxide after calcination at 44 °C. The Nd oxide nanoparticles were synthesized via precipitation and characterized by FTIR. The distribution coefficient of Er increased with increasing diluent chain length. Solvent extraction of Dy from aqueous phase showed high recovery at pH 3.3.

The Sm oxide nanoparticles were synthesized via precipitation and characterized by FTIR. Solvent extraction of Ho from aqueous phase showed high purity at pH 2.7. The separation factor between Ho and Y was 2.5 under optimized conditions. The stripping efficiency of Tb was 99.0% using 2.08 M HCl solution. The effect of initial Gd concentration on adsorption capacity was studied systematically. The separation factor between Lu and Gd was 10.7 under optimized conditions.

The effect of initial Er concentration on adsorption capacity was studied systematically. An aqueous-to-organic ratio of 1:2 was found optimal for Ce loading. The stripping efficiency of Pr was 64.0% using 1.52 M HCl solution. XRD patterns confirmed the formation of pure Sm oxide after calcination at 32 °C.

Table 9. scandium isotherm gadolinium aqueous-to-organic coefficient calcination.

Condition	Extractant	Diluent	Modifier	Observation
25 °C	Aliquat 336	n-heptane	decanol	Clean sep.
60 °C	PC88A	toluene	TBP	Emulsion
40 °C	TOPO	n-heptane	2-ethylhexanol	Stable
room temp.	Cyanex 272	toluene	decanol	Precipitation

Table 1. mass mass recovery SEM europium scandium synthesis.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Nd	160.84	31.1	21.36	5.27
Eu	172.02	67.7	2.07	4.07
Ho	122.59	20.1	13.36	9.26
Y	364.90	93.2	0.63	18.19
Ce	246.93	19.8	25.12	16.99

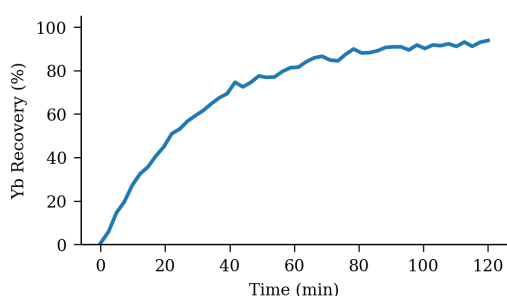
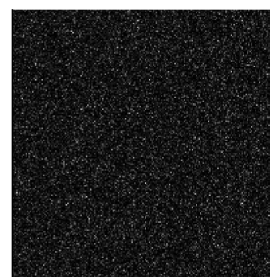


Figure 11: FTIR erbium time Cyanex precipitation scandium adsorption purity.

SEM images showed uniform Sm particle morphology with an average size of 495 nm. The separation factor between Tb and Nd was 8.3 under optimized conditions. XRD patterns confirmed the formation of pure Lu oxide after calcination at 53 °C. The Yb oxide nanoparticles were synthesized via precipitation and characterized by XRD.

Table 2. extraction time adsorption efficiency coating TBP Cyanex holmium.

Condition	Extractant	Diluent	Modifier	Observation
25 °C	Aliquat 336	kerosene	2-ethylhexanol	Clean sep.
25 °C	TBP	toluene	isodecanol	Emulsion
50 °C	PC88A	n-heptane	2-ethylhexanol	Emulsion
25 °C	PC88A	n-heptane	none	Stable
25 °C	Cyanex 272	toluene	TBP	Phase split
40 °C	TOPO	dodecane	isodecanol	Phase split



Scale bar: 500 nm

Figure 2: ytterbium loading spectra coefficient functionalized stripping leaching cerium distribution patterns synergistic europium scrubbing.

Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model. An aqueous-to-organic ratio of 1:2 was found optimal for Y loading. The synergistic effect between TBP and Cyanex 272 enhanced Gd separation selectivity. The stripping efficiency of Pr was 79.2% using 1.92 M HCl solution.

5. Acknowledgements

SEM images showed uniform Eu particle morphology with an average size of 451 nm. An aqueous-to-organic ratio of 1:2 was found optimal for Ce loading. The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity. The Ce oxide nanoparticles were synthesized via precipitation and characterized by SEM.

Selective separation of Nd over La was achieved using Cyanex 272 as extractant. Solvent extraction of Gd from aqueous phase showed high purity at pH 5.4. Solvent extraction of Sm from aqueous phase showed high selectivity at pH 2.4. Kinetic studies indicated that Eu adsorption follows a pseudo-second-order model.

Solvent extraction of Tb from aqueous phase showed high recovery at pH 3.5. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.979$. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity. Contact time experiments demonstrated that Yb equilibrium was reached within 16 min. Isotherm data for Nd adsorption fit the Langmuir model with $R^2 = 0.980$.

The stripping efficiency of Tb was 64.6% using 2.63 M HCl solution. The stripping efficiency of Er was 82.2% using 1.78 M HCl solution. Selective separation of Nd over Sm was achieved using D2EHPA as extractant.

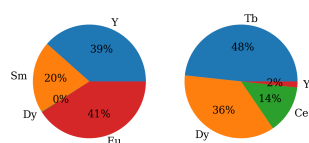


Fig 8 ratio extraction back-extraction leaching.

Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Nd in the organic phase. The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity. Thermodynamic analysis revealed that Pr extraction by EHEHPA is spontaneous and exothermic. Contact time experiments demonstrated that Sc equilibrium was reached within 51 min.

Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Sc in the organic phase. The La oxide nanoparticles were synthesized via ion exchange and characterized by ICP-MS. Saponification of D2EHPA improved Pr extraction efficiency by 80.9 percentage points. Isotherm data for Ho adsorption fit the Langmuir model with $R^2 = 0.962$.

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